IT 462: BIG DATA SYSTEMS DIABETES HEALTH INDICATORS

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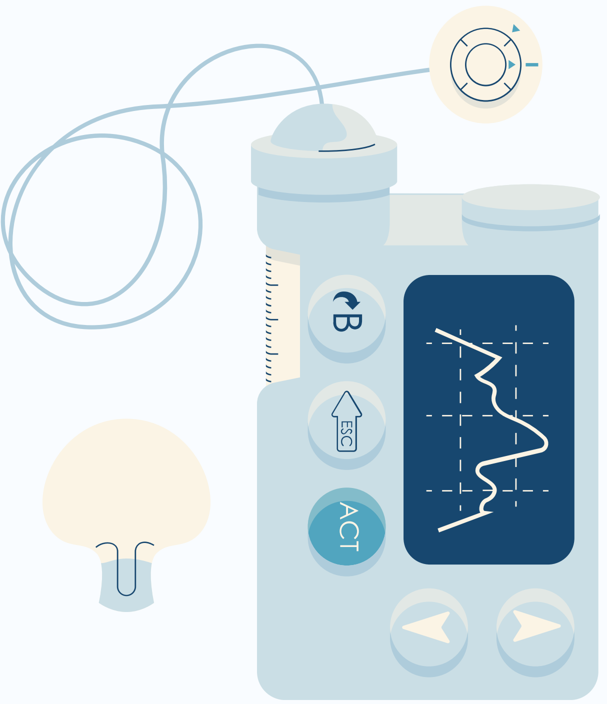
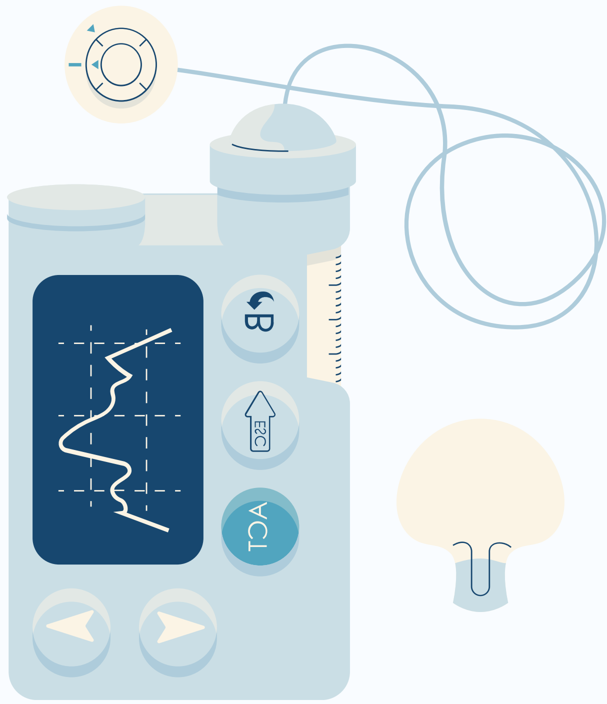
PROBLEM & DATASET



The Problem Millions remain undiagnosed with diabetes until serious complications arise. Can we predict diabetes risk using only survey data without any lab tests, nor clinical visits?

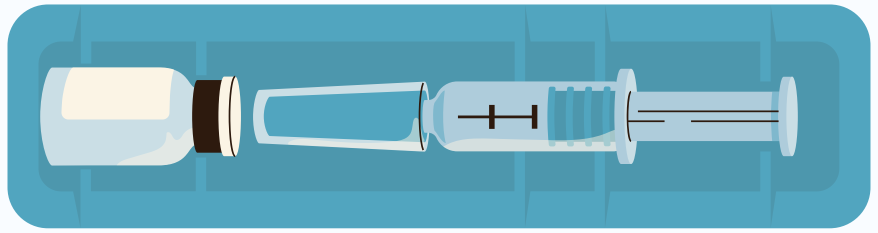
The Dataset

- **Source:** CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015
- **229,500 records** · 22 variables · ~22 MB CSV
- **Target:** Diabetes_binary (0 = No Diabetes, 1 = Prediabetes, 2= Diabetes)



Why It Matters Early identification using lifestyle and demographic data enables cheaper, wider screening, especially in underserved communities.

PREPROCESSING & EXPLORATION



Main Challenges

- **Class imbalance:** majority of records are non-diabetic
- **Mixed variable types:** binary, ordinal, and continuous features

Interesting Patterns Found Early

- **Diabetes rate rises sharply with age:** ~7% at age 18–24 → over 40% at 80+
- **Over 77%** of the data population is **overweight** or **obese**
- **22%** of respondents self-report **Fair** or **Poor** general **health**

RDD HIGHLIGHTS

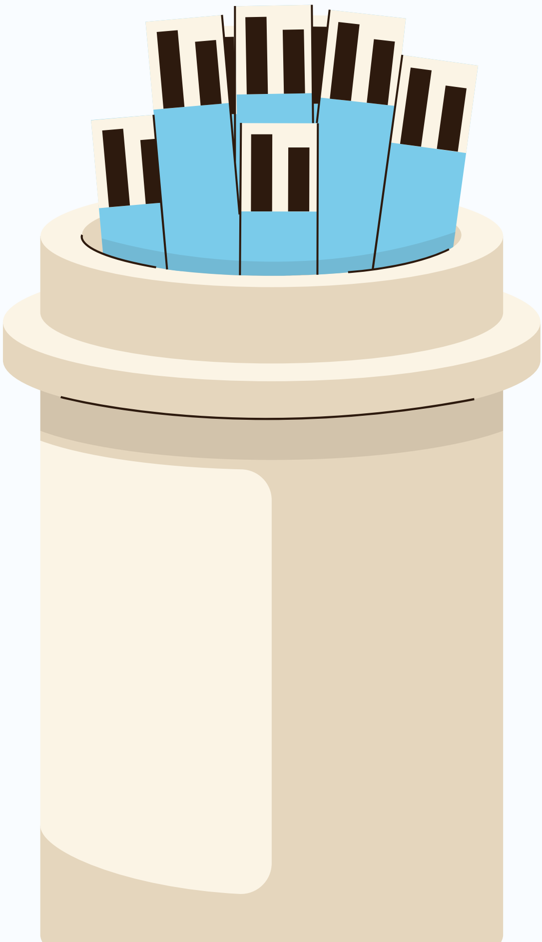
Key RDD Operations Used

map · filter · reduceByKey · mapValues · sortBy · sortByKey

Insight 1: Age & Diabetes Diabetes rate climbs steadily with age group. Age is the single clearest demographic predictor.

Insight 2: High-Risk Clustering 14,823 individuals (6.46%) simultaneously have high BP, smoking history, physical inactivity, and poor self-rated health risk factors cluster together, not appear in isolation.

Insight 3: BMI Distribution 41% of the dataset falls in the Obese category (BMI ≥ 30). Average BMI: 31.86 for diabetic vs 27.43 for non-diabetic individuals.

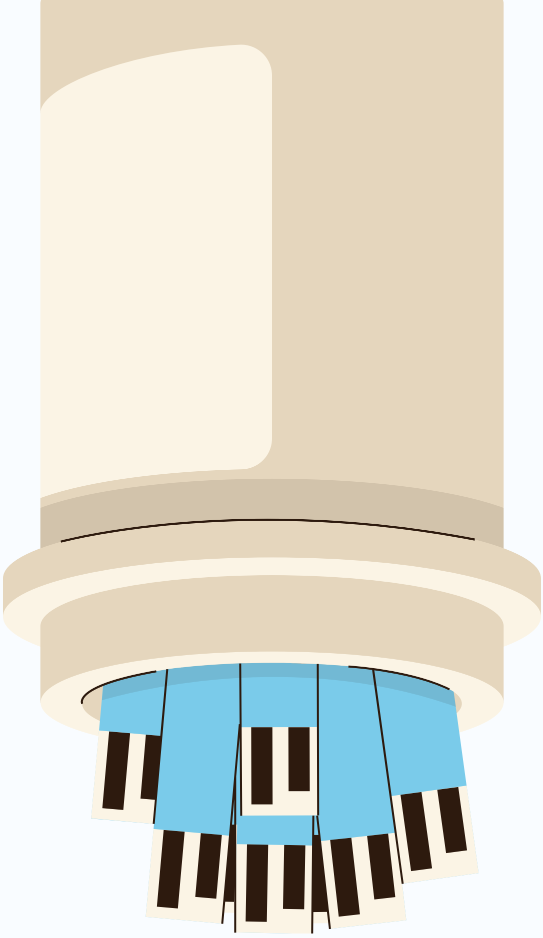


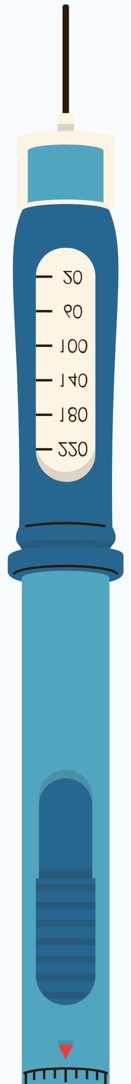
SQL HIGHLIGHTS

5 Analytical Queries Run with Spark SQL

Query	Key Finding
Diabetes Rate by Age Group	Rate rises from 1.80% (Age 1) to 25.82% (Age 11)
Avg BMI by Diabetes Status	Non-diabetic: 27.96 · Prediabetes/Diabetic: >30
High-Risk Individual Count	3,823 with 4+ combined risk factors
Top Age Brackets by Prevalence	Age codes 10–13 consistently highest
BMI Category Distribution	Obese: 84,661 · Overweight: 82,859

SQL confirmed what RDD exploration suggested: age and BMI are the dominant structured predictors of diabetes risk in this dataset.





MACHINE LEARNING: SETUP

Task: Multi-class classification

- Class 0: No Diabetes
- Class 1: Prediabetes
- Class 2: Diabetes

Pipeline Stages Raw Features → StringIndexer → VectorAssembler → RandomForestClassifier → Predictions

Why Random Forest?

- Handles mixed variable types well
- Robust to noise and nonlinear relationships
- Produces feature importance scores for interpretability

Hyperparameters

Trees: 50 · Max Depth: 10 · Seed: 42

Split: 70% train / 30% test

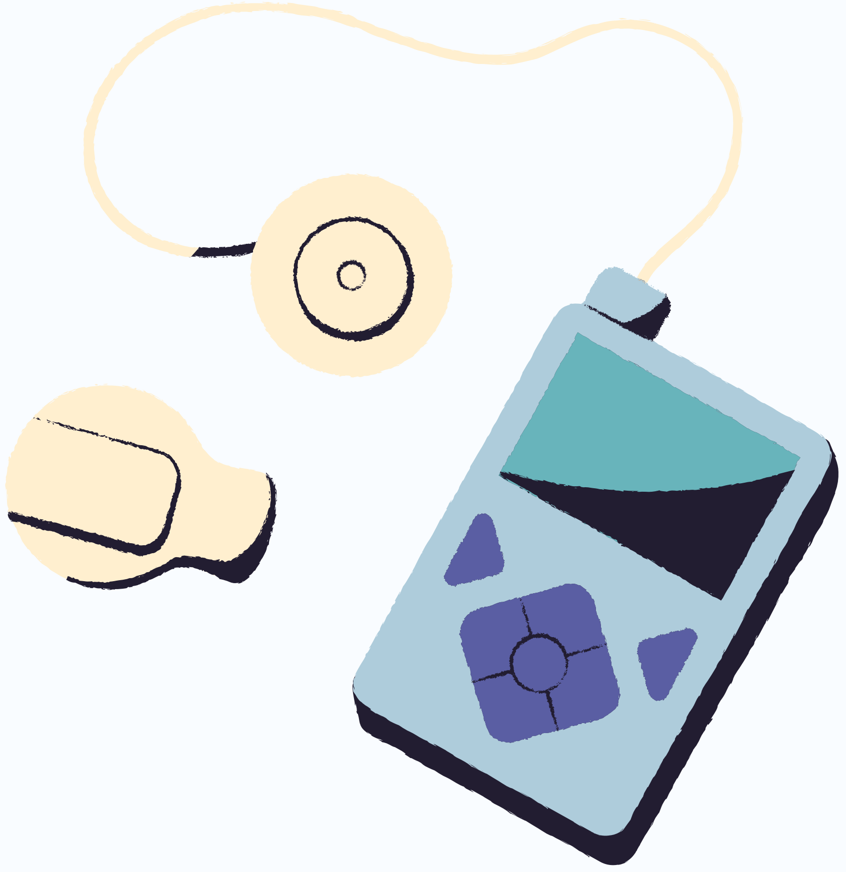
MACHINE LEARNING: RESULTS

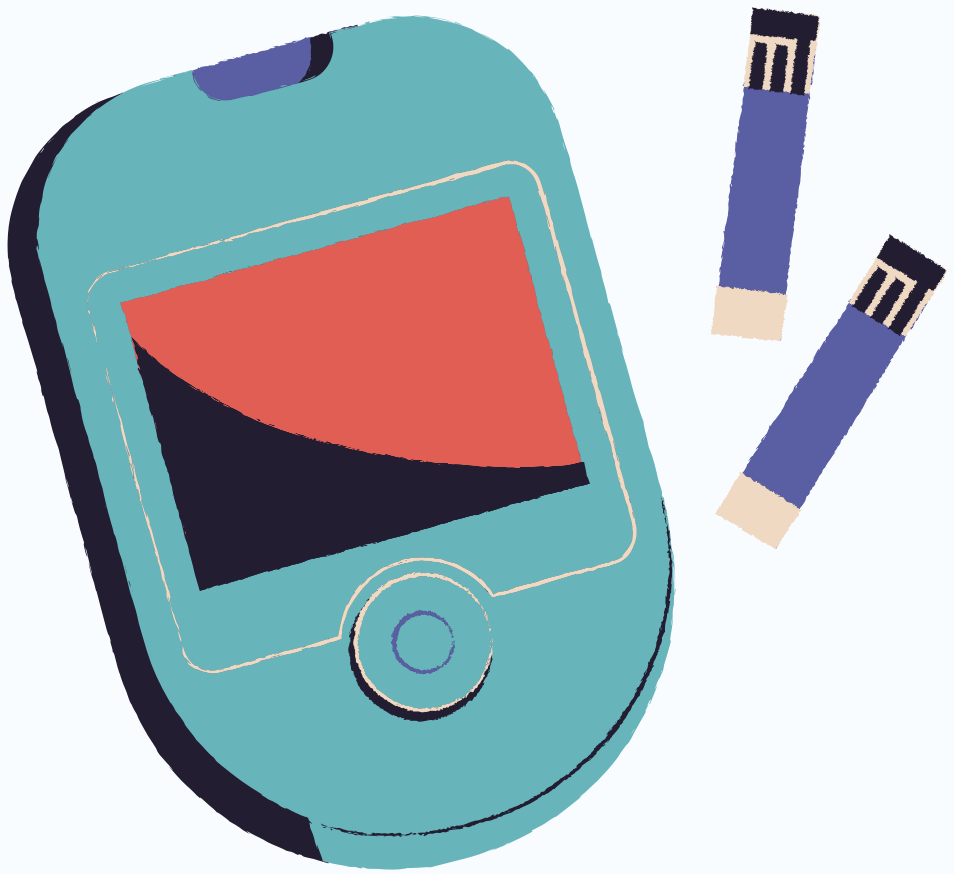
Model	Accuracy	F1-Score	Baseline Accuracy
Random Forest	83.49%	78.28%	82.82%

What this means: The model beats the baseline, but the gap is narrow class imbalance keeps inflating the baseline. Accuracy alone is misleading here; F1-score of 78.28% better reflects real predictive power across all three classes.

Top Predictors (**Feature Importance**)

1. General Health (GenHlth)
2. Blood Pressure (HighBP)
3. Body Mass Index (BMI)
4. Age
5. Income





THANK YOU

ANY QUESTIONS?